A discrete Lyapunov certificate for an aggressive weak-gradient iteration

Abstract

We verify a discrete Lyapunov estimate for the aggressive weak-gradient iteration of Ushiyama, Sato and Matsuo. The method and its energy are inherited. For a globally smooth strongly convex potential, an exact one-step identity gives an explicit sufficient step interval, geometric energy decay and weighted dissipation. The energy controls both original phase coordinates despite its negative quadratic term. Consequently every independent initial phase converges geometrically to the equilibrium. The upper-step equality and equal smoothness and convexity constants are included. The proof supplies the discrete coefficient calculation and initialization scope for this printed map; it makes no maximal-range or method-priority claim.

Keywords: discrete dynamical system; Lyapunov function; inertial iteration; global stability; strong convexity

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1 The iteration and its energy

Ushiyama, Sato and Matsuo [1, Section 3.1, p. 1337] introduce the aggressive iteration denoted there by wDGagr_sc. They display an energy and omit the proof of its nonincrease. They also leave a theoretical convergence estimate for this method unspecified. We give a sufficient step interval and an exact dissipation identity for that literal explicit map. The resulting conclusion concerns both coordinates of the original discrete phase.

Let $d \geq 1$ be finite, let $0 < \mu \leq L < \infty$, and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable, μ-strongly convex, with globally L-Lipschitz gradient. Thus

$$\begin{aligned} f(u) &\geq f(y) + \langle \nabla f(y), u - y \rangle + \frac{\mu}{2}\|u - y\|^2, \\ \|\nabla f(u) - \nabla f(y)\| &\leq L\|u - y\| \qquad (u, y \in \mathbb{R}^d). \end{aligned} \tag{1}$$

Write x^* for the unique minimizer and $f^* = f(x^*)$. Fix $h > 0$ and put $a = h\sqrt{\mu}$. The iteration is

$$\begin{aligned} z_k &= \frac{(1+a)x_k + av_k}{1+2a}, \\ v_{k+1} &= \frac{v_k + 2az_k - \dfrac{9a}{4\mu}\nabla f(z_k)}{1+2a}, \\ x_{k+1} &= \frac{x_k + av_{k+1}}{1+a}, \qquad k \geq 0. \end{aligned} \tag{2}$$

The initial variables $x_0, v_0 \in \mathbb{R}^d$ are independent. The variable v_k is the auxiliary phase coordinate of the printed method; it is not being identified with the physical velocity of another continuous model.

For completeness, (2) is obtained from the aggressive display in [1, Section 3.1] by choosing

$$\overline{\nabla} f(y,x) = \nabla f(x), \qquad (\alpha,\beta,\gamma) = (L/2, \mu/2, 0) \tag{3}$$

and solving its linear coupling. These constants are those of [1, Definition 1 and Proposition 1(i)]. In particular, the query in (2) is at z_k. No adjustable weak-gradient constant or nonlinear implicit solve remains after (3) is fixed.

For $X = x - x^*$ and $V = v - x^*$, define the inherited energy

$$E(x,v) = f(x) - f^* + \frac{4\mu}{9}\left(\|V\|^2 - \frac{1}{2}\|V - X\|^2\right). \tag{4}$$

It is exactly the energy in the aggressive-method display of [1]. Its continuous antecedent is due to Moucer, Taylor and Bach [3]. The quadratic term alone can be negative. Strong convexity nevertheless makes the whole energy coercive on the original phase, as proved below.

Put

$$\bar{h} = \frac{2}{\sqrt{9L - 4\mu} - 2\sqrt{\mu}}, \qquad q = 1 + 2a. \tag{5}$$

The denominator in (5) is positive, including $L = \mu$.

Theorem 1. *Under* (1)*, the map* (2) *is globally well defined for every* $h > 0$*. The energy* (4) *is continuous and proper, is nonnegative, and vanishes exactly at* (x^*, x^*)*. If*

$$0 < h \le \bar{h}, \tag{6}$$

then every initial phase satisfies, with $E_k = E(x_k, v_k)$*,*

$$qE_{k+1} \le E_k - \frac{\mu a}{9}\|z_k - x^*\|^2 - \frac{\mu a^2}{18q^2}\|v_k - x_k\|^2. \tag{7}$$

In particular,

$$E_k \le q^{-k} E_0, \tag{8}$$

$$\sum_{k=0}^{\infty} q^k \left(\frac{\mu a}{9}\|z_k - x^*\|^2 + \frac{\mu a^2}{18q^2}\|v_k - x_k\|^2\right) \le E_0, \tag{9}$$

$$f(x_k) - f^* \le 9q^{-k} E_0, \tag{10}$$

$$\begin{aligned} \|x_k - x^*\|^2 + \|v_k - x^*\|^2 &\le \frac{63}{\mu} q^{-k} E_0 \\ &\le \frac{63L}{2\mu} q^{-k}\left(\|x_0 - x^*\|^2 + \|v_0 - x^*\|^2\right). \end{aligned} \tag{11}$$

Thus (x^*, x^*) *is globally exponentially stable in the iteration index. All assertions include* $h = \bar{h}$*,* $L = \mu$*, and the zero-energy initial state.*

The step restriction is structural: it depends only on μ, L and not on the initial energy or dimension. It is sufficient, and no maximality is asserted. The proof uses the displayed method and energy, standard strong-convexity and smoothness inequalities, and a particular exact quadratic identity. General systematic Lyapunov frameworks for fixed-step methods already exist [4]; neither that framework nor a new iteration is a contribution of this note.

2 Coercivity before the trajectory argument

Strong convexity gives a coercive affine-quadratic lower bound for f. Since f is continuous on finite-dimensional space, its minimum is attained; strong convexity makes it unique, and $\nabla f(x^*) = 0$. The denominators in (2) are positive. Computing z_k, its ordinary gradient, v_{k+1} and then x_{k+1} therefore gives a unique finite next phase. No bounded-orbit hypothesis is needed for this construction.

Lemma 1. *For every phase (x, v), with $X = x - x^*$ and $V = v - x^*$,*

$$E(x,v) \geq \frac{f(x) - f^*}{9} + \frac{2\mu}{9}\|V + X\|^2, \tag{12}$$

$$E(x,v) \geq \frac{\mu}{18}\|X\|^2 + \frac{2\mu}{9}\|V + X\|^2, \tag{13}$$

$$\frac{\mu}{63}\left(\|X\|^2 + \|V\|^2\right) \leq E(x,v) \leq \frac{L}{2}\left(\|X\|^2 + \|V\|^2\right). \tag{14}$$

Proof. Completing the square in (4) gives

$$E = f(x) - f^* - \frac{4\mu}{9}\|X\|^2 + \frac{2\mu}{9}\|V + X\|^2.$$

Since $f(x) - f^* \geq \mu\|X\|^2/2$, this proves (12) and (13). Also

$$\|X\|^2 + \|V\|^2 \leq 3\|X\|^2 + 2\|V + X\|^2 \leq \frac{63}{\mu}E.$$

The smoothness bound at x^* gives $f(x) - f^* \leq L\|X\|^2/2$. Dropping the negative square in (4) and using $4\mu/9 \leq L/2$ proves the upper bound in (14). These arguments do not use the step restriction. The lower bound, continuity and finite dimensionality imply that energy sublevels are compact. Equation (13) also identifies the zero set exactly. ∎

3 An exact one-step identity

We normalize only for the algebra. Set

$$\Phi(U) = \frac{f(x^* + U) - f^*}{\mu}, \qquad \ell = \frac{L}{\mu} \geq 1, \qquad b = 1 + a, \quad q = 1 + 2a.$$

Then $\Phi(0) = 0$, $\nabla\Phi(0) = 0$, Φ is 1-strongly convex and its gradient is ℓ-Lipschitz. For one step put

$$X = x - x^*, \quad V = v - x^*, \quad Z = z - x^*, \quad Y = x_+ - x^*, \quad W = v_+ - x^*, \quad G = \nabla\Phi(Z).$$

Define

$$Q(X,V) = \frac{2}{9}\|V\|^2 + \frac{4}{9}\langle V, X\rangle - \frac{2}{9}\|X\|^2, \qquad e(X,V) = \Phi(X) + Q(X,V). \tag{15}$$

Thus $E(x,v) = \mu e(X,V)$. Introduce the nonnegative curvature residuals

$$S_X = \Phi(X) - \Phi(Z) - \langle G, X - Z\rangle - \frac{1}{2}\|X - Z\|^2, \tag{16}$$

$$S_L = \Phi(Z) - \Phi(Y) + \langle G, Y - Z\rangle + \frac{\ell}{2}\|Y - Z\|^2, \tag{17}$$

$$S_0 = \langle G, Z\rangle - \frac{1}{2}\|Z\|^2 - \Phi(Z). \tag{18}$$

Strong convexity at Z, with endpoints X and 0, gives $S_X, S_0 \geq 0$. The Lipschitz-gradient bound gives $S_L \geq 0$. This last assertion follows by integrating $\nabla\Phi$ on the segment from Z to Y; no Hessian is assumed.

Lemma 2. *For every phase and every $a > 0$, put*

$$D = V - X, \qquad T = D + \frac{9q}{4}G, \qquad \Theta = 4 + 8a + (8 - 9\ell)a^2. \tag{19}$$

The literal map (2) *satisfies the exact equality*

$$\begin{aligned} e(X,V) - qe(Y,W) = {} & \frac{a}{9}\|Z\|^2 + \frac{a^2}{18q^2}\|D\|^2 + \frac{a^2\Theta}{18b^2q^3}\|T\|^2 \\ & + S_X + qS_L + 2aS_0. \end{aligned} \tag{20}$$

Proof. Direct substitution into (2) gives

$$\begin{aligned} &X = Z - \frac{a}{q}D, \qquad V = Z + \frac{b}{q}D, \\ &Y = Z - \frac{a^2}{bq^2}T, \quad W = Z + \frac{1}{q}D - \frac{a}{q^2}T, \quad G = \frac{4}{9q}(T - D). \end{aligned} \tag{21}$$

These identities impose no relation between X and V. By (16)–(18) and $q - 1 = 2a$,

$$\begin{aligned} \Phi(X) - q\Phi(Y) = {} & \langle G, X - qY\rangle + a\|Z\|^2 + \frac{1}{2}\|X - Z\|^2 - \frac{q\ell}{2}\|Y - Z\|^2 \\ & + S_X + qS_L + 2aS_0. \end{aligned} \tag{22}$$

Expanding the quadratic form (15) after (21) gives

$$\begin{aligned} \langle G, X - qY\rangle + Q(X,V) - qQ(Y,W) = {} & -\frac{8a}{9}\|Z\|^2 - \frac{4a^2}{9q^2}\|D\|^2 \\ & + \frac{2a^2(2a^2 + 2a + 1)}{9b^2q^3}\|T\|^2. \end{aligned} \tag{23}$$

In particular, the mixed products $\langle Z, D\rangle$, $\langle Z, T\rangle$ and $\langle D, T\rangle$ cancel. This calculation uses only bilinearity, so it is valid in every dimension.

To see the remaining coefficients explicitly, use

$$\|X - Z\|^2 = \frac{a^2}{q^2}\|D\|^2, \qquad \|Y - Z\|^2 = \frac{a^4}{b^2q^4}\|T\|^2.$$

Adding (23) to (22), the coefficients of $\|Z\|^2$ and $\|D\|^2$ become $a/9$ and $a^2/(18q^2)$. The coefficient of $\|T\|^2$ is

$$\frac{2a^2(2a^2 + 2a + 1)}{9b^2q^3} - \frac{\ell a^4}{2b^2q^3} = \frac{a^2[4 + 8a + (8 - 9\ell)a^2]}{18b^2q^3}.$$

This proves (20) as an equality, before any step restriction is imposed. ∎

Let $\eta = \sqrt{9\ell - 4}$. Since $\ell \geq 1$, one has $\eta \geq \sqrt{5} > 2$, and

$$\Theta = [2 + (2 - \eta)a][2 + (2 + \eta)a]. \tag{24}$$

For $a > 0$, the second factor is positive. Thus $\Theta \geq 0$ on exactly the interval

$$0 < a \leq \frac{2}{\eta - 2}.$$

Using $a = h\sqrt{\mu}$ shows that this is (6). Dropping the nonnegative last square and curvature residuals in (20) and multiplying by μ gives (7). This proves descent before any boundedness or limiting statement about the orbit is used.

At $h = \bar{h}$, only the coefficient of $\|T\|^2$ becomes zero. The two retained coefficients in (7) remain positive. Their squared terms vanish simultaneously only if $Z = D = 0$, which implies $X = V = 0$ by (21). When $L = \mu$, the bound is

$$\bar{h} = \frac{4 + 2\sqrt{5}}{\sqrt{\mu}}.$$

No division by $L - \mu$ or Θ has been used, so this endpoint is included. The factorization in (24) determines the sign of this particular certificate. It does not prove necessity of the step interval for energy decrease or phase convergence.

4 The original phase and weighted dissipation

Define

$$d_k = \frac{\mu a}{9}\|z_k - x^*\|^2 + \frac{\mu a^2}{18q^2}\|v_k - x_k\|^2 \geq 0.$$

The proved inequality is $qE_{k+1} \leq E_k - d_k$. Since Lemma 1 already gives $E_k \geq 0$, induction yields (8). For every finite N, multiplication by q^k and telescoping gives

$$\sum_{k=0}^{N} q^k d_k \leq E_0 - q^{N+1}E_{N+1} \leq E_0.$$

The nonnegative partial sums are increasing and bounded, proving (9). No limit of $q^{N+1}E_{N+1}$ is required for this conclusion.

Apply (12) and (14) at time k, and then the upper bound in (14) at time zero. This gives (10) and (11). In the product Euclidean norm, the latter reads

$$\|(x_k - x^*, v_k - x^*)\| \leq \sqrt{\frac{63L}{2\mu}}\, e^{-k\log(q)/2}\|(x_0 - x^*, v_0 - x^*)\|.$$

Since $q > 1$, this is global exponential stability for each fixed admissible step. If $E_0 = 0$, the initial phase is (x^*, x^*). Since $\nabla f(x^*) = 0$, (2) then leaves it fixed. No division by initial energy or any vector norm occurs. This completes the proof of Theorem 1.

5 Relation to the existing estimates

The weak-gradient inequality in [1, Definition 1] is obtained in the present case by combining the smooth upper bound and the strongly convex lower bound at the same query z. Our three residuals retain those inequalities separately. The squared-norm identity (23) then determines the coefficient (24) for the aggressive coefficients of the printed map. No new interpolation principle is needed.

The standard weak-gradient theorem in [2, Theorem 5.5 and Appendix E.4] uses different update coefficients and the energy

$$f(x) - f^* + (\beta + \gamma)\|v - x^*\|^2.$$

Its step restriction is therefore not a premise of Theorem 1. In particular, the comparison scale h_0 used for the standard method in [1, Section 3.2] is not imported as an aggressive-method bound. The present result does not evaluate or refute the numerical implementation or Figure 1 of that paper; its conclusion is for the explicit formula (2) and every independent initial phase.

The continuous energy antecedent [3] allows the quadratic part to be indefinite while requiring positivity of the complete Lyapunov expression. Lemma 1 applies precisely that elementary distinction to the printed energy (4). A continuous-time decay estimate alone does not prove the discrete identity (20). Similarly, membership in the fixed-step framework of [4] does not supply a particular certificate without its coefficient calculation.

The source comparison uses the 2024 version of record, the official NeurIPS 2023 paper, and the current author version arXiv:2205.12772v2 for the relevant continuous statement in [3]. The full SIAM journal body and every OpenReview discussion were not inspected. No correction to the relevant displays was found in the checked official records, but this is not an exhaustive priority search. The note supplies an explicit sufficient Lyapunov and trajectory estimate for an existing method, with no claim of a new method, new energy, optimal rate, maximal step range or firstness. The accompanying technical note records the exact source and normalization distinctions.

References

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